

Q1. Build the Playground with Wildcards

```
rushi@rushi: ~ /lab

rushi@rushi:~$ cd lab/
rushi@rushi:~/lab$ touch {a1,a2,b1}.txt
rushi@rushi:~/lab$ touch {a3,b2}.log
rushi@rushi:~/lab$ touch {notes,readme}.md
rushi@rushi:~/lab$ touch data.csv
rushi@rushi:~/lab$ ls *txt
a1.txt  a2.txt  b1.txt
rushi@rushi:~/lab$ ls -n a
ls: cannot access 'a': No such file or directory
rushi@rushi:~/lab$ ls *md
notes.md  readme.md
rushi@rushi:~/lab$ ls a*
a1.txt  a2.txt  a3.log
rushi@rushi:~/lab$ rm *log
rushi@rushi:~/lab$ ls *.log 2>/dev/null || echo none
none
rushi@rushi:~/lab$ ls *txt | wc -l
3
rushi@rushi:~/lab$
```

Q2. Chaining with Pipes.

```
rushi@rushi:~/lab$ ls /etc | wc -l
231
rushi@rushi:~/lab$ ls /etc | head -n 10
adduser.conf
alsa
alternatives
anacrontab
apache2
apg.conf
apm
apparmor
apparmor.d
appport
rushi@rushi:~/lab$ ls /etc | sort -r
zsh_command_not_found
xml
xdg
xattr.conf
X11
wpa_supplicant
wgetrc
vulkan
```

```
bindresvport.blacklist
bash_completion.d
bash_completion
bash.bashrc
avahi
apt
appstream.conf
apport
apparmor.d
apparmor
apm
apg.conf
apache2
anacrontab
alternatives
alsa
adduser.conf
rushi@rushi:~/lab$ date | wc -w
7
rushi@rushi:~/lab$ ls /etc | grep -i "conf" | wc -l
41
rushi@rushi:~/lab$
```

Q3. Searching Inside Files with grep

```
rushi@rushi:~/lab$ echo "Alice Engineer 30" > people.txt
rushi@rushi:~/lab$ echo "Bob Manager 45" >> people.txt
rushi@rushi:~/lab$ echo "Carol Engineer 28" >> people.txt
rushi@rushi:~/lab$ echo "Dave Designer 38" >> people.txt
rushi@rushi:~/lab$ echo "Eve Engineer 41" >> people.txt
rushi@rushi:~/lab$
rushi@rushi:~/lab$ grep "Engineer" people.txt
Alice Engineer 30
Carol Engineer 28
Eve Engineer 41
rushi@rushi:~/lab$ grep -n "Engineer" people.txt
1:Alice Engineer 30
3:Carol Engineer 28
5:Eve Engineer 41
rushi@rushi:~/lab$ grep -c "Engineer" people.txt
3
rushi@rushi:~/lab$ grep -v "Engineer" people.txt
Bob Manager 45
Dave Designer 38
rushi@rushi:~/lab$ grep "^Cpeople.txt
rushi@rushi:~/lab$ grep "C" people.txt
Carol Engineer 28
rushi@rushi:~/lab$ grep -i "engineer" people.txt
Alice Engineer 30
Carol Engineer 28
Eve Engineer 41
rushi@rushi:~/lab$
```

Q4. Hunting Files With find

```
rushi@rushi:~/lab$ find *txt
a1.txt
a2.txt
b1.txt
people.txt
rushi@rushi:~/lab$ find -type d
.
rushi@rushi:~/lab$ mkdir -p deep/hidden.txt
rushi@rushi:~/lab$ find *txt
a1.txt
a2.txt
b1.txt
people.txt
rushi@rushi:~/lab$ find . *txt
.
./data.csv
./people.txt
./notes.md
./a2.txt
./b1.txt
./readme.md
./deep
./deep/hidden.txt
./a1.txt
a1.txt
a2.txt
b1.txt
people.txt
rushi@rushi:~/lab$ find -name '*.txt'
./people.txt
./a2.txt
./b1.txt
./deep/hidden.txt
./a1.txt
rushi@rushi:~/lab$ find -name 'a*'
./a2.txt
./a1.txt
rushi@rushi:~/lab$ find -name '*.txt' | wc -l
5
rushi@rushi:~/lab$
```

Q5. The real power :combine them

```
rushi@rushi:~/lab$ find -name "*.txt" -c
find: unknown predicate `-c'
rushi@rushi:~/lab$ find -c -name "*.txt"
find: unknown predicate `-c'
rushi@rushi:~/lab$ find -name "*.txt" | wc -l
5
rushi@rushi:~/lab$ find -name "*.txt" | wc -c
58
rushi@rushi:~/lab$ find -name "*.txt" | -c
-c: command not found
rushi@rushi:~/lab$ find -name "*.txt" | grep -c
Usage: grep [OPTION]... PATTERNS [FILE]...
Try 'grep --help' for more information.
rushi@rushi:~/lab$ find -name "*.txt" | grep -c .
5
rushi@rushi:~/lab$
```

```
rushi@rushi:~/lab$ find -name "*.txt" | wc -l
5
rushi@rushi:~/lab$ grep "Engineer" people.txt | sort
Alice Engineer 30
Carol Engineer 28
Eve Engineer 41
rushi@rushi:~/lab$ grep "conf" /etc | head -n 5
grep: /etc: Is a directory
rushi@rushi:~/lab$ ls /etc | grep "conf" | head -n 5
adduser.conf
apg.conf
appstream.conf
brltty.conf
ca-certificates.conf
rushi@rushi:~/lab$ awk '{print $2}' people.txt | sort | uniq -c
    1 Designer
    3 Engineer
    1 Manager
rushi@rushi:~/lab$ awk '{print $2}' people.txt | sort | uniq -c | sort -rn
    3 Engineer
    1 Manager
    1 Designer
rushi@rushi:~/lab$
```

Q6. Saving the Result

```
rushi@rushi:~/lab$ grep "Engineer" people.txt | sort > engineers.txt

rushi@rushi:~/lab$ ls
a1.txt  b1.txt    deep          notes.md    readme.md
a2.txt  data.csv  engineers.txt people.txt

rushi@rushi:~/lab$ cat engineers.txt
Alice Engineer 30
Carol Engineer 28
Eve Engineer 41

rushi@rushi:~/lab$ date >> engineers.txt
rushi@rushi:~/lab$ wc -l engineers.txt
4 engineers.txt

rushi@rushi:~/lab$ find -name "*.txt"
./people.txt
./engineers.txt
./a2.txt
./b1.txt
./deep/hidden.txt
./a1.txt

rushi@rushi:~/lab$ find -name "*.txt" > all-text-files.txt
rushi@rushi:~/lab$ tail -n 3 all-text-files.txt
./b1.txt
./deep/hidden.txt
./a1.txt
rushi@rushi:~/lab$
```

Q7. The Investigator

```
rushi@rushi:~/lab$ echo "GET ok" > access.log
rushi@rushi:~/lab$ echo "GET ok" >> access.log
rushi@rushi:~/lab$ echo "GET ok" >> access.log
rushi@rushi:~/lab$ echo "GET ok" >> access.log
rushi@rushi:~/lab$ echo "GET ok" >> access.log
rushi@rushi:~/lab$ echo "POST ok" >> access.log
rushi@rushi:~/lab$ echo "POST ok" >> access.log
rushi@rushi:~/lab$ echo "POST ok" >> access.log
rushi@rushi:~/lab$ echo "GET fail" >> access.log
rushi@rushi:~/lab$ echo "GET fail" >> access.log
rushi@rushi:~/lab$ grep "fail" access.log | wc -l
2
rushi@rushi:~/lab$ sort access.log | uniq -c
    2 GET fail
    5 GET ok
    3 POST ok
rushi@rushi:~/lab$ sort access.log | uniq -c | sort -rn
    5 GET ok
    3 POST ok
    2 GET fail
rushi@rushi:~/lab$ grep "^GET" access.log | wc -l
7
```